

MELFA RV-1A/RV-2AJ



**Combining Compact Design & Operational Ease.
Mitsubishi Electric's newest all-purpose industrial
robots designed for the industrial world.**

Debut

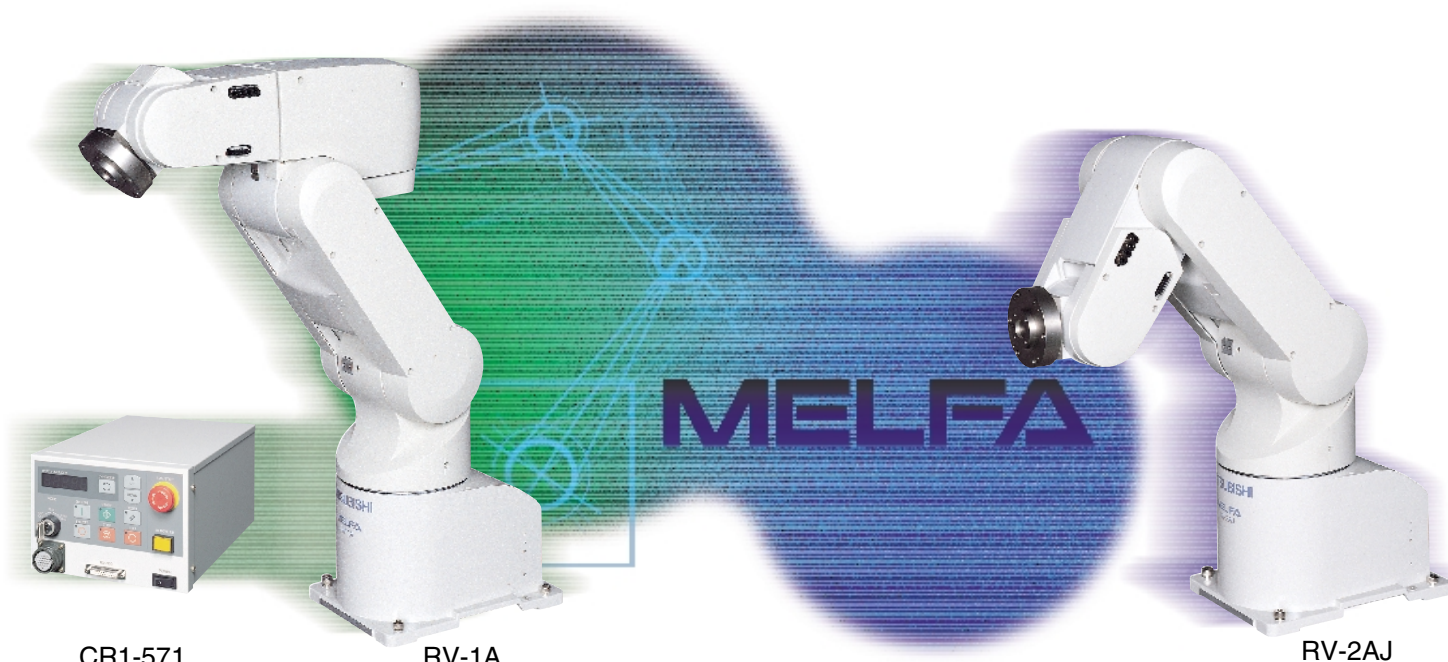
Mitsubishi Electric Industrial Robots are manufactured at a factory certified for ISO14001 (standards for environmental management systems) and ISO9001 (standards for quality assurance management systems).



World's Most Wanted Industrial Robots Debut

MELFA All-purpose Industrial

Mitsubishi Electric has redesigned the world's best selling robot, the RV-M1, into a high-performance, all-purpose industrial robot that offers faster processing, greater accuracy and open network capability. When it comes to improving production facilities with compact, high-precision equipment, there's nothing else like it.



CR1-571

RV-1A

RV-2AJ



Compact Controller

Robot controller the size of an 8½×11 sheet of paper



Compact, Light-weight 6-axis Unit

Simple installation, Simple layout



All-purpose Industrial Use

64 bit controller & new mechanism



Environmental Safety

Energy-saving overall motor maximum capacity of only 200W

Basic Specifications

Main Body

Model		RV-1A	RV-2AJ
Motion degrees-of-freedom		6	5
Motor drive		AC servomotor (J1-J3 and J5 axis brake attached)	
Position detection		Absolute encoder	
Maximum load capacity (rated)*1		kg	kg
Arm length		mm	mm
Maximum radial reach		mm	mm
Operating limits	J1	±150	
	J2	180 (–60~+120)	
	J3	95 (+60~+155)	230 (–110~+120)
	J4	±160	—
	J5	±90	
	J6	±200	
Maximum speed	J1	180	
	J2	90	
	J3	135	
	J4	180	—
	J5	180	
	J6	210	
Maximum quasi-speed		mm/s	mm/s
Repeat position accuracy		mm	mm
Weight		kg	kg
Tool wiring*2		Input: 4 points (hand) Output: 4 points (base)	
Tool air piping		ø4×4 poles (base–hand)	
Installation		Floor-standing, hanging from ceiling	

*1: Maximum load capacity is that with wiring flange surface in downward position (±10°).

*2: Pneumatic hand interface (optional) is required when tool output is used.

*3: Three slots when extension option box (optional) is attached.

*4: Electricity consumption is approximately 0.2kW during general pattern operation.

*5: Default is set to 180~253 VAC at the time of shipping (CE specifications is set to 230 VAC±10% at the time of shipping).

*6: Size or weight excluding attachment of extension option box.

Controller

Model		CR2-571
Path control		PTP, CP
No. of control axes		axis
CPU		64 bit RISC/DSP
Main functions		Indirect interpolation, direct interpolation, 3-dimensional radi interpolation, palletizing, condition branching, subroutines, multi-tasking, optimum adjustable speed control, optimum override control, optimum route connection, etc.
Programming language		MELFA-BASIC IV and MoveMaster Language (MoveMaster Command)
Position learning		Teaching method, MDI method
Memory capacity	Teaching position	Points
	No. of steps	Steps
	No. of programs	Points
External input	General I/O	Points
	Exclusive input	Points
	Hand I/O	Points
	Emergency stop input	Points
	Door switch input	Points
	RS232C	Ports
Interface	RS422	Ports
	Exclusive hand slot	Slots
	Extension slot*3	Slots
	Robot I/O link	Channels
	Temperature	°C
	Humidity	%RH
Power supply	Input voltage limit	V
	Power capacity*4	kVA
	Ground	Ω
Configuration		Independent mounting, open configuration
Overall dimensions*6		mm
Weight*6		kg

Note: Specifications are subject to change without prior notification.

Robots

Features



Compact, Lightweight Body

The MoveMaster takes up minimal space and is easy to install. A main body weight of less than 20kg makes the robot easy to carry or relocate and the space needed for controller installation is the size of an 8 1/2x11 sheet of paper, the smallest in this robot category (company comparison).



High-performance All-purpose Industrial Robot

Equipped with AC servomotors and absolute encoders, the robot offers a shorter tact time with a maximum quasi-speed of 2.1m/s. Furthermore, the combination of an amazingly high accuracy of +0.02mm with 6-axis operation, gives the new MoveMaster the capability to perform complicated tasks.



Improved Network Functions

The incorporation of a 64 bit CPU enhances network capabilities for Ethernet and CC-Link. Connectivity to high-performance PCs and PLCs and vision sensors, enables higher processing speeds and increased productivity.



Energy Savings and Safety

Use of an AC servomotor reduces each axis output to less than 50W and total motor output to less than 200W, creating energy savings for plant operations. Internal packaging of wiring and piping has been incorporated to improve safety.

MoveMaster Compatibility

The MoveMaster uses MELFA-BASIC IV and MoveMaster Language (MoveMaster Command) as standard robot programming languages. As the same arm length, installation base and hand attachment dimensions as the RV-M1 MoveMaster are used, replacement and/or exchange of robot types are simplified.

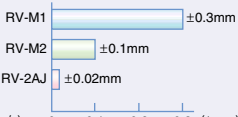
RV-M1 Comparison (controller)



Maximum Quasi-speed Comparison



Repeat Position Accuracy Comparison



RV-M1 Comparison (main body)

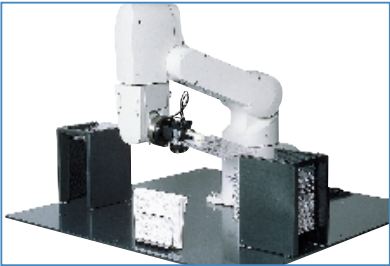


(previous model)
RV-M1

RV-2AJ

Main Applications

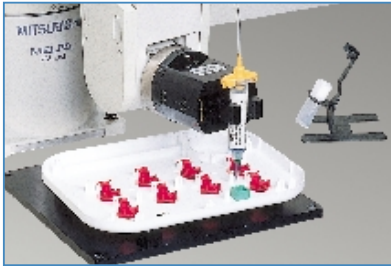
High-precision Electronics Machinery Industry



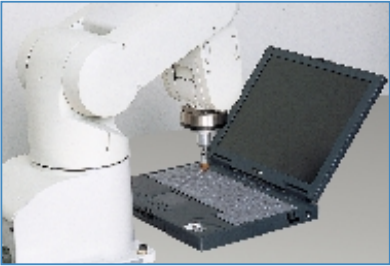
CD-ROM management/transfer



Assembly of electric machinery parts, etc.



Adhesive application and sealing



Keyboard inspection



Inspection/Testing of electronic parts



Inspection/Analysis of medical products

Education, Research, Amusement Industries



Instructional aids for education and R&D

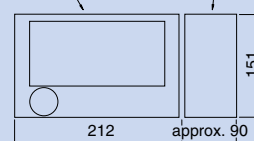


Amusement and advertisement demonstrations

Foods Industry

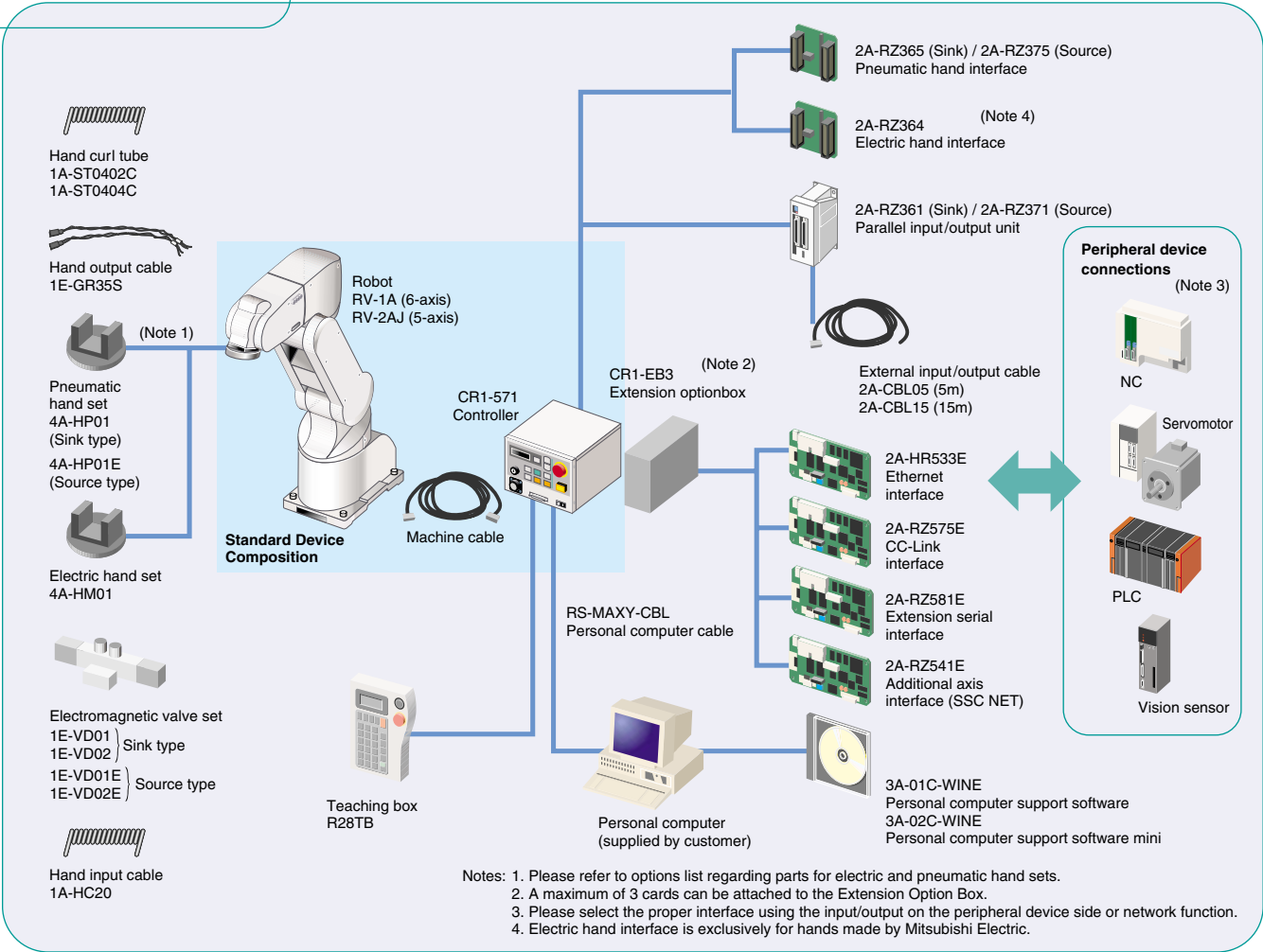


Packaging of foods



Organization of the RV-1A and RV-2AJ

System Configuration



Options

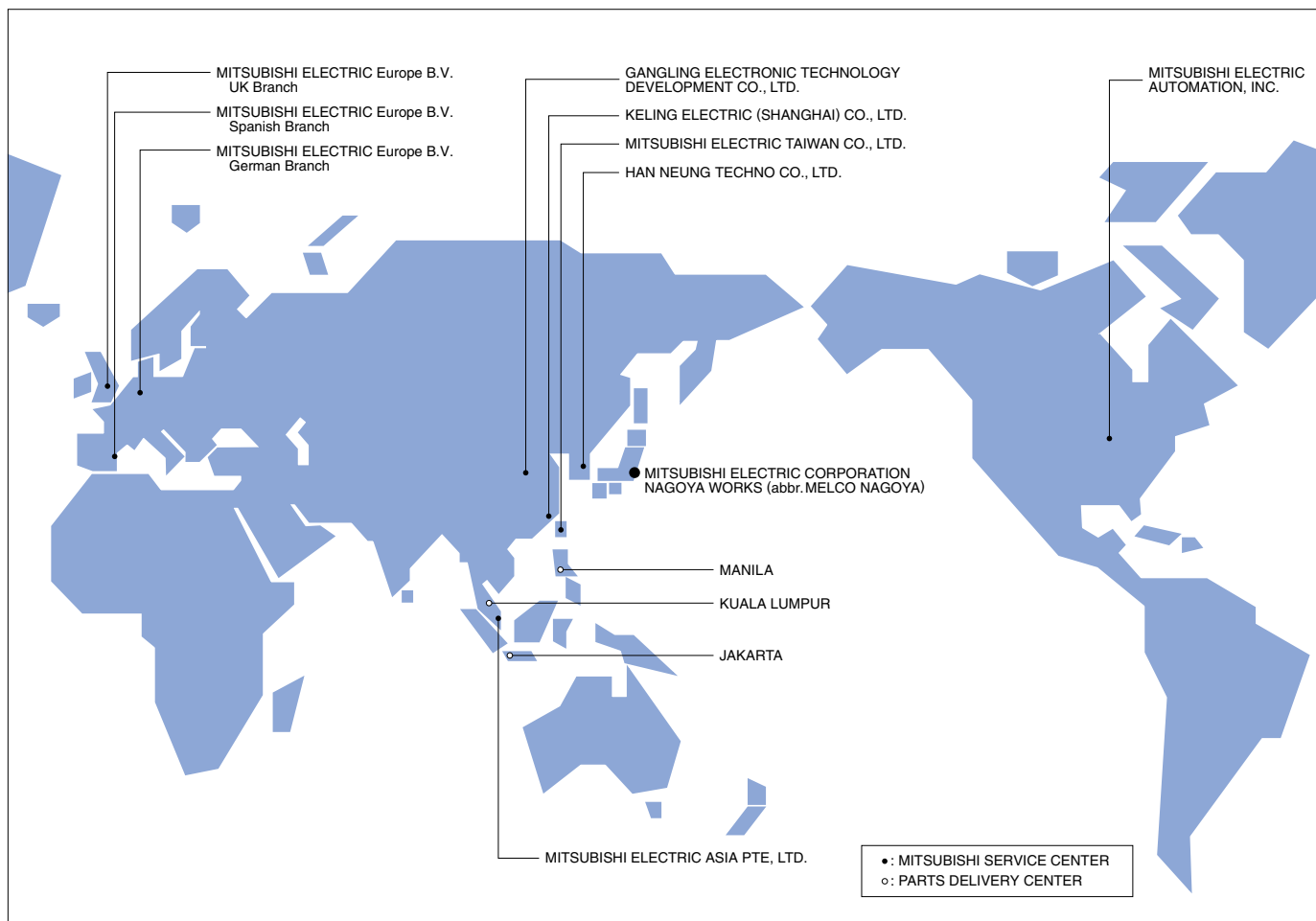
	Name	Model No.	Description/Specification	Remarks
Main body options	Electric hand set	4A-HM01	Main hand unit, curl cable, electric hand interface, adapter, installation bolts	
	Pneumatic hand set	4A-HP01	Main hand unit, curl tube (1 set), pneumatic hand interface, electromagnetic valve (1 set), adapter, installation bolts	Sink type
	Electromagnetic valve set	4A-HP01E		Source type
		1E-VD01	1 set	Sink type
		1E-VD02	2 sets	
		1E-VD01E	1 set	Source type
		1E-VD02E	2 sets	
	Hand output cable	1E-GR35S	350mm length	
	Hand input cable	1A-HC20	200mm length	
	Hand curl tube (1 set: 2 pieces)	1A-ST0402C	External diameter $\phi 4 \times 2.5\text{mm}$	
Controller options	Hand curl tube (2 set: 4 pieces)	1A-ST0404C	External diameter $\phi 4 \times 2.5\text{mm}$	
	Hand adapter	1A-HA01	for conversion of flange attached to mutual hand of RV-M1	
	Teaching box	R28TB	IP65, cable length 7m	
	Pneumatic hand interface (sink type) / (source type)	2A-RZ365 / 2A-RZ375	Output: 8 points (sink) / (source)	
	Parallel Input/Output interface (sink type) / (source type)	2A-RZ361 / 2A-RZ371	Output: 32 points, Input: 32 points (sink) / (source)	
	External Input/Output cable	2A-CBL05	One terminal untreated, 5m length	
		2A-CBL15	One terminal untreated, 15m length	
	Ethernet interface	2A-HR533E	10base-T 10Mbps	
	CC-Link interface	2A-RZ575E	CC-Link intelligent remote station (1 or 4 stations), 32 points/32points per station	Attached to extension box
	Extension serial interface	2A-RZ581E	RS-232C/422, 1 channel each (422 can be switched to 232C), I/F 2-channel encoder	
Maintenance	Additional axis interface	2A-RZ541E	for SSC NET (Max. 8 axes) *1	
	Extension option box	CR1-EB3	Option card extension unit, 3 slots	Note 1
	Personal computer support software (Windows®)	3A-01C-WINE	Windows® compatible support software, CD-ROM	
	Personal computer support software mini (Windows®)	3A-02C-WINE	Simplified version of Windows® compatible support software, CD-ROM	
	Personal computer cable	RS-MAXY-CBL	for PC-AT (DOS/V) compatible systems, 3m length	
	Backup battery	A6BAT	for internal mechanisms	
		ER6	for internal controller	

*1: Connect with only MR-J2 and MR-J2 super

Note 1: Required when *1 option is used.

* Windows® is a registered trademark of Microsoft Corporation.

OVERSEAS SERVICE NETWORK & GLOBAL SERVICE NETWORK



MITSUBISHI ELECTRIC CORPORATION NAGOYA WORKS (MELCO NAGOYA)

1-14 Yada-minami 5-chome, Higashi-ku, Nagoya,
Aichi prefecture, Japan
Telephone 81-52-712-2212/Fax 81-52-712-3692

MITSUBISHI ELECTRIC AUTOMATION, INC. (Chicago Headquarters)

500 Corporate Woods Parkway, Vernon Hills, IL 60061, U.S.A.
Telephone 1-847-478-2311/Fax 1-847-478-2253

MITSUBISHI ELECTRIC Europe B.V. German Branch
Gothaer Strasse 10, 40880 Ratingen, Germany
Telephone 49-2102-486-334/Fax 49-2102-486-717

MITSUBISHI ELECTRIC Europe B.V. UK Branch
Travellers Lane, Hatfield, Hertfordshire, AL10 8XB, UK
Industrial Automation Systems Division
Telephone 44-1707-276100/Fax 44-1707-278695

MITSUBISHI ELECTRIC Europe B.V. Spanish Branch
Pol. Ind. "Can Magi" - C/. Joan Buscallà, 2-4 - A.C. 420
08190 Sant Cugat del Vallès (Barcelona)
Telephone 34-935-653-168/Fax 34-935-891-579

HAN NEUNG TECHNO CO., LTD.

Dongseo Game Channel Bldg. 2F. 660-11,
Deungchon-Dong Kangseo-Ku
Seoul, 157-030, Korea
Telephone 82-3663-0493/Fax 82-2-3663-0475

MITSUBISHI ELECTRIC TAIWAN CO., LTD. (Taipei FA Center)

3th Fl., No.122, WU KUNG 2nd Road, WU-KU Hsiang,
Taipei Hsien, Taiwan
Telephone 886-2-2299-3060/Fax 886-2-2298-1909

(Taichung FA Center)

No. 8-1, Gong Yeh 16th Road, Taichung Industrial Park,
Taichung City, Taiwan
Telephone 886-4-359-0688/Fax 886-4-359-0689

MITSUBISHI ELECTRIC ASIA PTE, LTD.

307 Alexandra Road #05-01/02
Mitsubishi Electric Building Singapore 159943
Telephone 65-473-2308/Fax 65-476-7439

GANGLING ELECTRONIC TECHNOLOGY DEVELOPMENT (BEIJING) CO., LTD.

Room 954, Office Building, New Century Hotel
No. 6 Southern Road, Capital Gym., Beijing, 100044 China
Telephone 86-10-6849-2077/Fax 86-10-6849-2087

KELING ELECTRIC (SHANGHAI) CO., LTD.

2F Block 5 Building Automation Instrumentation Plaza 103
Cao Bao Rd. Shanghai 200233 China
Telephone 86-21-6484-9360/Fax 86-21-6484-9361



Safety Warning

To ensure proper use of the products listed in this catalog,
please be sure to read the instruction manual prior to use.



MITSUBISHI ELECTRIC CORPORATION

HEAD OFFICE: MITSUBISHI DENKI BLDG., 2-2-3, MARUNOUCHI, CHIYODA-KU, TOKYO 100-8310, JAPAN